STAR MF 2.0

Webhook

BSE STAR MF 2.0 – Webhook Sequence Document

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1. Document Summary

Document Title	Webhook Sequence Flow – All Modules
System Name	STAR MF 2.0 – Member Integration
Purpose	Defines webhook sequences, callback structures, and acknowledgment expectations for all modules integrated with STAR MF 2.0.

2. Webhook Architecture Overview

The STAR MF 2.0 system communicates updates to member systems using HTTP POST webhooks. Each module (UCC, Mandate, Order, SIP, Payment, etc.) has its own payload structure and event triggers. Webhooks are used for sending status updates, verification results, processed order details, and triggering next-level actions.

Module Action → STAR MF 2.0 Server → Webhook POST → Member Endpoint → Member sends 200 OK

3. Understanding Event Types

The **Event Type** defines the source module of each webhook (UCC, Mandate, Order, SIP, or Payment) and directs how the member system should interpret and handle the received payload.

Modules	Event Type
UCC – Client On-Boarding	UCC
Order Manager	ORDER
Systematic Orders Engine	SXP
Mandates	MANDATES
Payments Manager	PAYMENT GATEWAY

4. Reading Sequential Status Flow

- Each event corresponds to a specific status code (Starting from 0 for initial stage).
- Status Code help determine whether the process flow currently stands.
- You can refer to status flow table in each module, to interpret stage transitions.
- **S** as status code indicates Special Scenarios.

5. Modules Classification

5.1. UCC (Client Onboarding)

This module handles client registration and verification by generating a **Unique Client Code** (UCC), performing **PAN validation**, executing **KYC compliance checks**, and conducting **bank verification**. For **Demat** clients, **DP (Depository Participant)** verification is performed. Upon successful completion of all checks, the client is **activated** for **transactions**. Each webhook event corresponds to a specific stage in the **onboarding** workflow.

Physical Clients (2FA Included Members) Onboarding: -

Event	Event Flow Code	Description
PENDING_AUTHENTICATION	0	This will be the initial event triggered (Client authentication pending)
ucc_auth_ucc	1	This event will be triggered when, client authentication is accepted by the Client.
PENDING_VERIFICATION	2	This event indicates that the client's details are undergoing third party verifications (PAN/KYC/Bank etc.).
ucc_elog_verification	2	This event will be triggered (Client Elog authentication is pending)
ucc_kyc_verification	3	This event indicates UCC KYC Verification is Completed.
ucc_pan_verification	3	This event indicates UCC Pan Verification is Completed.
ucc_bank_acc	3,5	This event indicates UCC Bank Verification is Completed.
ucc_auth_elog	3	Its status is contingent upon user acceptance and can be triggered at any stage of the workflow cycle after (ucc_elog_verification)
ucc_auth_nominee	3	Its status is contingent upon user acceptance and can be triggered at any stage of the workflow cycle after (PENDING_VERIFICATION)
ACTIVE	4	The event indicates the UCC is active and available for Transaction
suspended	S	This event will only trigger if the UCC has been suspended by Exchange.

Note: For members with 2FA enabled, all related webhooks will be triggered based on the services activated and any exemptions opted for the member code.

Physical Clients (2FA excluded Members) Onboarding:

Event	Event Flow Code	Description
PENDING_VERIFICATION	0	This event indicates that the client's details are undergoing third party verifications (PAN/KYC/Bank etc.).
ucc_kyc_verification	1	This event indicates UCC KYC Verification is Completed.
ucc_pan_verification	1	This event indicates UCC Pan Verification is Completed.
ucc_bank_acc	1,3	This event indicates UCC Bank Verification is Completed.
ACTIVE	2	The event indicates the UCC is active and available for Transaction
suspended	S	This event will only trigger if the UCC has been suspended by Exchange.

Demat Clients Onboarding:

Event	Event Flow Code	Description
PENDING_VERIFICATION	0	This event indicates that the client's details are undergoing third party verifications (PAN/KYC/Bank etc.).
ucc_kyc_verification	1	This event indicates UCC KYC Verification is Completed.
ucc_pan_verification	1	This event indicates UCC Pan Verification is Completed.
ucc_dp_acc	1	This event indicates UCC DP Verification is Completed.
ucc_bank_acc	1,3	This event indicates UCC Bank Verification is Completed.
ACTIVE	2	The event indicates the UCC is active and available for Transaction
suspended	S	This event will only trigger if the UCC has been suspended by Exchange.

Active UCC (Transaction Ready)

- For a Physical UCC, submission of the AOF or E-Log is mandatory, the UCC will remain inactive until these details are submitted.
- Bank verification does not affect UCC activation.
- For 2FA-enabled members with nomination, the UCC will remain inactive until nominee authentication is not completed by the client.

Call Back Structure (UCC):


```
{
 "member": {
  "member_id": "XXXX",
  "subbr_code": "",
  "subbr_arn": "",
  "euin": "",
  "euin_flag": "",
  "partner_id": ""
 },
 "request_id": "d645bd5a-cded-4cea-a172-273aa916ac65",
 "investor": {
  "client_code": "XXXX"
 },
 "parent_client_code": "",
 "action": {
  "msgcode": 10001,
  "at": "2025-07-07 05:31:05.698221877 +0530 IST m=+57981.238208414",
  "event_type": "UCC",
  "event": "PENDING_VERIFICATION",
  "event_message": "",
  "data": null
 }
}
```


5.3 Order Manager (ORDERS)

Handles the complete order lifecycle (Lumpsum/Switch & Redemption) for BSE Star MF transactions, including order creation, validation, submission, and status tracking. It ensures seamless coordination between client applications and BSE systems, managing both Lumpsum and SIP orders with real-time updates and reconciliation through integrated webhooks.

Order Manager – Physical Flow

Lumpsum

Event	Event Flow Code	Description
received	0	Order Received
order_2fa_pending	1	Order 2FA Pending - In case of 2FA included member
bank_tpv_pending	2	Bank TPV Pending
payment_pending	2	Payment Pending
match_pending	3	OMB Pending - ICCL
matched	4	OMB Completed - ICCL
sent_to_rta	5	Final CFF File reported to RTA
queued_for_rta	6	CFF in RTA Processing
rta_reprocess	7	In case of CFF Upload level failure - If Order need to be reprocessed
rta_resp_rcvd	7	RFF Recived from RTA
rta_rejected	7	Order Rejected by RTA
units_rta_settled	8	Units Settled by RTA
done	9	Order Cycle Complete
platform_rejected	S	This event will be triggered if order placed exceeds threshold limit.
threshold_approval_pending	S	This event will be triggered if threshold is in approval stage.
ops_rejected	S1	This event will be triggered if threshold is not approved by Ops team

Redemption

Event	Event Flow Code	Description
received	0	Redemption Order Placed
order_2fa_pending	1	In Case of 2FA enabled Member
sent_to_rta	2	CFF Reported to RTA
queued_for_rta	3	CFF in RTA Processing
rta_resp_rcvd	4	IF Valid RFF Received from RTA
rta_rejected	4	If Invalid RFF is received from RTA or order rejected through UI
redempt_rta_settled	5	Redemption Completed
done	6,7	Redemption Order Completed
amc_mis_payout_updated	6,7	Payout report recived from AMC

Switch –

Event	Event Flow Code	Description
received	0	Switch Order Received
order_2fa_pending	1	In Case of 2FA enabled Member
queued_to_rta	2	In Process - BSE <> RTA
send_to_rta	3	CFF Reported to RTA
sw_rta_so_valid	4	IF Valid Switch Out RFF is received
rta_rejected	4	If Invalid Switch OUT RFF is received from RTA or order rejected through UI
sw_rta_si_valid	5	IF Valid Switch In RFF is received
rta_rejected	5	If Invalid Switch In RFF is received from RTA or order rejected through UI
done	6	Switch Order Completed

Notes:

- For 2FA-exempt members, the order_2fa_pending event will not be triggered.
- For BSE PG Service enabled members, the order events will be synchronized with the payment events. In a lumpsum transaction, the flow will be **payment_pending** (order event) -> **payment gateway events** -> **match_pending** (ORDER event).
- For SXP orders, specifically in the case of XSIP, the order events will start with **sxp_order_triggered -> match_pending** -> **order events flow.**
- For SXP Orders, incase of SIP transactions, without mandate the events will begin with **sxp_order_triggered** -> **payment_pending** -> **payment gateway events** -> **match_pending** -> **order events flow.**
- "**received**" will be the initial event, and "**done**" will be the final event for all successful transaction processes, including lumpsum, switch, and redemption.
- For BSE PG Service excluded members, no payment-related events will be triggered; the order events will proceed directly from **payment_pending** to **match_pending.**

Order Manager – Demat Flow

Lumpsum

Event	Event Flow Code	Description
received	0	Order Received
bank_tpv_pending	1	Bank TPV Pending
payment_pending	2	Payment Pending
match_pending	3	OMB Pending - ICCL
matched	4	OMB Completed - ICCL
sent_to_rta	5	Final CFF File reported to RTA
queued_for_rta	6	Final CFF File queued to RTA (if rejection is not received in receipt-in) CFF in RTA Processing
rta_reprocess	7	In case of CFF Upload level failure - If Order need to be reprocessed
rta_resp_rcvd	7	RFF Received
rta_rejected	7	Order Rejected by RTA
queued_for_dp	8	Order Queued to DP
dep_rejected	9	Order Rejected by DP
dp_units_matched	9	Units' allotment done
units_payout_sent	10	Units Payout sent
done	11	Order Cycle Complete
dep_rejected	12	In Case of Units bounce
units_return_to_amc	13	Units Retured to AMC

Lumpsum – In Case of Partial Units Allocation

Event	Event Flow Code	Description
queued_for_dp	8	Order Queued to DP
dep_rejected	9	Order Rejected by DP
partial_units_matched	9	Partial Units' allotment done
units_payout_sent	10	Partial units Payout sent
queued_for_dp	11	Order Queued to DP
dp_units_matched	12	Complete Units allotment Done
units_payout_sent	13	Units Payout sent
done	14	Order Cycle Complete
dep_rejected	15	In Case of Units bounce
units_return_to_amc	16	Units Retured to AMC

Redemption

Event	Event Flow Code	Description
received	0	Redemption order Received
cff_validation_pending	1	Validation CFF reported to RTA
rta_rejected	2	Rejection Received in CFF Validation
units_pool_init	3	Units pooling initiated
units_rcpt_in_sent	4	Receipt_in File Sent to DP
units_pull_completed	5	Units pooling completed
dep_rejected	5	Rejection Received in Receipt-in Response
sent_to_rta	6	Final CFF File reported to RTA
queued_for_rta	7	Final CFF File queued to RTA (if rejection is not received in receipt-in) (CFF in RTA Processing)
rta_resp_rcvd	8	IF valid RFF is received
rta_rejected	8	IF invalid RFF is received or Order is rejected through UI
amc_mis_pending	9	If Valid RFF is Received
red_matched	10	IF MIS - VALID RFF Details & Bank receipt is matched
redempt_payout_attempted	11	PAYOUT attempt to UCC (N/5 N- No of bank accounts available in UCC.
done	12	If Payout is success than order is done
redempt_payout_failed	11	If payout is failed
redempt_fund_returned_amc	12	If the attempt are exhausted then funds is return back to AMC

Switch

Event	Event Flow Code	Description
received	0	Demat Switch Order received
cff_validation_pending	1	Validation CFF file reported to RTA
rta_rejected	2	Rejection is received in CFF file
units_pool_init	3	Units Pooling Initiated
units_rcpt_in_sent	4	Rcpt In file send to DP
units_pool_completed	5	Units Pooling Completed
dep_rejected	5	Rejection Received in Receipt-in Response
sent_to_rta	6	CFF File sent to RTA (if rejection is not received in receipt-in)
queued_for_rta	7	Final CFF File queued to RTA (if rejection is not received in receipt-in) (CFF in RTA Processing)

sw_rta_so_valid	8	IF Valid Switch Out RFF is received
rta_rejected	8	If Invalid Switch OUT RFF is received from RTA or order rejected through UI
sw_rta_si_valid	9	IF Valid Switch In RFF is received
rta_rejected	9	If Invalid Switch In RFF is received from RTA or order rejected through UI
queued_for_dp	10	If valid Switch-in RFF is received
dep_rejected	11	Order rejected by DP
dp_units_matched	11	If Switch-in units are received in payin file
units_payout_sent	12	Payout file is generated for switch-in transaction
done	14	switch order process

Refund

Exchange Refund

Event	Event Flow Code	Description
refund_pending	0	Refund Pending
exch_refund_init	1	Exchange Refund Initiated
exch_refund_attempted	2	Exchange Refund attempted (**N/5** N - No of banks available in UCC)
received_by_bank	3	Refund Amt Received by customer
exch_refund_failed	4	Exchange refund failed
done	4	Refund Process Completed
funds_to_unclaimed	5	In Case if Refund failed, then this status will be triggered

AMC Refund

Event	Event Flow Code	Description
rta_rejected	0	Order Rejected By RTA
refund_pending	1	Order Pending For Refund
amc_refund_matched	2	Refund amount matched **at AMC**
amc_refund_attempted	3	AMC Refund Attempted (N/5 N- No of bank account available in UCC)
received_by_bank	4	Refund Received By Bank
amc_refund_failed	5	AMC Refund Failed
done	5	Refund process completed
funds_to_unclaimed	5	In Case if Refund failed, then this status will be triggered

Call Back Structure (ORDER):


```
{
  "member": {
    "member_id": "XXXX"
  },
  "request_id": "9e5d5007-98f3-4aeb-8256-a39576828b1f",
  "investor": {
    "client_code": "XXXXXXXXXX"
  },
  "action": {
    "msgcode": 10001,
    "at": "2025-07-07 05:31:05.698221877 +0530 IST m=+57981.238208414",
    "event_type": "ORDER",
    "event": "match_pending",
    "event_message": "",
    "order_id": "XXXXXXXXX",
    "mem_ord_ref_id": "XXXXXXXXXXXXXXX"
  }
}
```


Call Back Structure (ORDER): In Case of Exchange Refund, bank_rcpt_id will be part of the call back structure.


```
{
  "member": {
    "member_id": "XXXX"
  },
  "request_id": "9e5d5007-98f3-4aeb-8256-a39576828b1f",
  "investor": {
    "client_code": "XXXXXXXXXX"
  },
  "action": {
    "msgcode": 10001,
    "at": "2025-07-07 05:31:05.698221877 +0530 IST m=+57981.238208414",
    "event_type": "ORDER",
    "event": "exch_refund_init",
    "event_message": "",
    "order_id": "XXXXXXXXX",
    "mem_ord_ref_id": "XXXXXXXXXXXXXXX",
    "bank_rcpt_id": "XXXXXXXXXXXXX"
  }
}
```

5.3 Systematic Order Engine (SXP)

This module manages recurring investments by registering SXPs and scheduling installments. Each webhook event represents a specific stage in the SXP lifecycle.

Event	Event Flow Code	Description
reg	0	This will be initial event indicating SXP has been successfully registered at BSE.
sxp_2fa_pending	1	This event will be triggered in case of 2FA included members.
active	1,2	This event indicates the registered SXP Order is active. **For 2FA Included Member** – This event will be triggered after client completes the 2FA authentication. **For 2FA Excluded Members** – The SXP order will be default in active state.
sxp_order_triggered	3	This event will be triggered when child orders are initiated for the registered SXPs.
paused	S	This event will be triggered, depending upon client initiation.
sxp_investor_canc	S	This event will be triggered, depending upon client initiation.
cancelled	S	This event will be triggered, if the SXP order is cancelled by the exchange.
mandate_unlink	S	This event will be triggered, depending upon client initiation.
autocancelled	S	This event will only trigger if three consecutive SIPs fail.
matured	S	This event will only trigger, if SXP order has matured.

Note:

- For Members with 2FA enabled, across all functionalities such as active, cancel, paused, and mandate_unlink, the event sxp_2fa_pending will be triggered, upon which another event will follow based on functionality initiated.
- The SXP child order events will be included within the Order module.
- SXP stands for all systematic transactions (XSIP, SIP, STP, SWP & Special Products)

Call Back Structure (SXP):


```
{
  "member": {
    "member_id": "63655"
  },
  "request_id": "20447022-d6f5-4404-b2ea-26dcf87683de",
  "investor": {
    "client_code": "U8c4b53fb2913459b835c25bee7444ced"
  },
  "action": {
    "msgcode": 10001,
    "at": "2025-10-01 16:07:06.713424439 +0530 IST m=+17803.332425421",
    "event_type": "SXP",
    "event": "reg",
    "event_message": "",
    "sxp_reg_num": "202505000000256"
  }
}
```


Call Back Structure (SXP): When an SXP child order is triggered, the Order ID and member_ord_ref_id will also be included in the callback structure. (Note – In case of SXP child orders triggered, will be system generated.)


```
{
 "member": {
  "member_id": "XXXX"
 },
 "request_id": "20ffd5eb-54b4-4cc6-b8cc-9be3059452d1",
 "investor": {
  "client_code": "XXXXXXXXX"
 },
 "action": {
  "msgcode": 10001,
  "at": "2025-10-01 16:07:06.713424439 +0530 IST m=+17803.332425421",
  "event_type": "SXP",
  "event": "sxp_order_triggered",
  "event_message": "",
  "sxp_reg_num": "XXXXXXXXXXXXXXX",
  "order_id": "XXXXXXXXXX",
  "mem_ord_ref_id": "XXXXXXXXXXXX-4"
 }
}
```

5.4 MANDATES

This module handles the registration, verification, and activation of payment mandates. It tracks mandate registration, and monitors approval or rejection. Webhook events indicate the current stage of the mandate lifecycle, enabling real-time updates and reconciliation with the client account.

Mandate – ENach

Event	Event Flow Code	Description
initiated	0	This event will be triggered, when mandate registration process is initiated.
investor_auth_awaited	1	Investor authentication Pending
active	2	This event will be triggered, when mandate is successfully active at agency.
rejected	2	This event will be triggered if mandate registration is not completed at agency.
cancelled	S	This event will trigger if mandate is cancelled by the investor.
completed	S	This event will trigger when mandate **end date = current date**

Mandate – Nach

Event	Event Flow Code	Description
initiated	0	This event will be triggered, when mandate registration process is initiated.
scan_upload_pending	1	This event will be triggered, when physical document submission is pending.
in_process_agency	2	This event will be triggered, when mandate is in processing at agency.
active		This event will be triggered, when mandate is active at agency.
rejected	2	This event will be triggered if mandate registration process is not completed at agency.
cancelled	S	This event will trigger if mandate is cancelled by the investor.

completed	S	This event will trigger when mandate **end_date = current date**
auto_rejected	S	This event will be triggered, if mandate is rejected by agency.
investor_auth_awaited	S	This event will be triggered in case of 2FA enabled members, in case investor wants to cancel the mandate.

Mandate – UPI Autopay

Event	Event Flow Code	Description
initiated	0	This event will be triggered, when mandate registration process is initiated.
in_process_agency	1	Investor authentication Pending
active	2	This event will be triggered, when mandate is successfully active at agency.
rejected	2	This event will be triggered if mandate registration process is not completed at agency.
cancelled	S	This event will trigger if mandate is cancelled by the investor.
completed	S	This event will trigger when mandate **end date = current date**
auto_rejected	S	This event will be triggered, if mandate is rejected by agency.
pre_debit _notification_ sent_successfully	S	This event will be triggered, when amount has successfully been pre debited from the account.
investor_auth_awaited	S	This event will be triggered in case of 2FA enabled members, in case investor wants to cancel the mandate.

Note:

- In case of 2FA enabled members, during mandate cancellation by investor **investor_auth_awaited** event will be triggered followed by **cancelled** event.

Call Back Structure (MANDATES):


```
{
  "member": {
    "member_id": "XXXXXX"
  },
  "request_id": "2c2dd2f2-29f4-4589-8f89-43256bb5f15d",
  "investor": {
    "client_code": "XXXXXXXXXXXXX"
  },
  "action": {
    "msgcode": 10001,
    "at": "2025-10-01 16:07:06.713424439 +0530 IST m=+17803.332425421",
    "event_type": "MANDATES",
    "event": "initiated",
    "event_message": "",
    "mandate_id": "XXXXXXXXX"
  }
}
```

5.5 Payments Manager (BSE PG Service)

Manages complete payment processing for BSE Star MF transactions, covering initiation, redirection, and status tracking across Net Banking, UPI, and Mandate modes. The integrated webhook provides real-time payment and order status notifications to client systems, ensuring secure, reliable, and synchronized communication with BSE platforms.

Event	Event Flow Code	Description
exch_init	0	This event will be triggered, when payment request have been initiated by Exchan
agency_pending	1	Payment Gateway: Payment request is sent to an Agency, waiting for response
agency_inprocess	2	Payment Gateway: Payment request is in process at agency's end
agency_approved	3	Payment Gateway: Payment is approved by an agency
agency_rejected	3	Payment Gateway: Payment is rejected by an agency
agency_cancelled	3	This Status will trigger, when none of the event triggers
agency_payment_complete	4	This Status will trigger, when the complete payment process is completed.

Note:

- Payments Event flow will be part of the order event flow, will be in synchronization to the orders events flow.
- For BSE PG Service excluded members, no payment-related events will be triggered; the order events will proceed directly from **payment_pending** to **match_pending.**
- The payment event sequence remains the same for both UPI and NetBanking transactions.
- In the case of payment by OTM, the order events will proceed directly from **payment_pending** to **match_pending.**

Call Back Structure (Payment Manager):


```
{
  "member": {
    "member_id": "XXXX"
  },
  "request_id": "0930f36b-faa6-4e06-8684-9a832bbfe62e",
  "investor": {
    "client_code": "XXXXXXXXXX"
  },
  "action": {
    "msgcode": 10001,
    "at": "2025-10-01 16:07:06.713424439 +0530 IST m=+17803.332425421",
    "event_type": "PAYMENT GATEWAY",
    "event": "agency_rejected",
    "event_message": "",
    "pg_order_ids": [
      XXXXXXXXX
    ]
  }
}
```